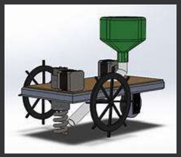
Automatic Seed Sowing Machine



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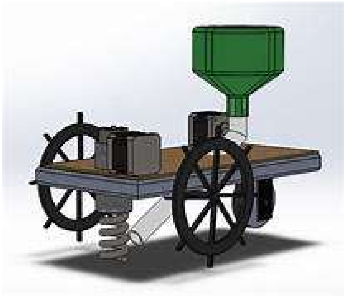
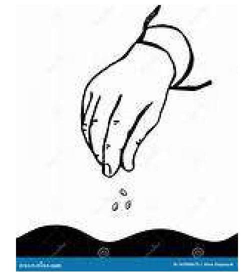
Abstract :

- The Automatic Seed Sowing Machine using IoT is a smart agricultural device designed to automate the seed sowing process, ensuring precise seed placement across large fields. Integrated with IoT sensors and a microcontroller, the machine monitors soil moisture, temperature, and other critical parameters, adjusting the sowing process in real-time for optimal seed germination. The GPS-enabled system navigates fields with accuracy, while a versatile seed dispenser ensures consistent spacing and reduces wastage.



Opportunity for Improvement:



Statement: As the global population is projected to reach nearly 10 billion by 2050, the agricultural sector must increase its efficiency to meet the rising demand for food. Small and medium sized farms, which form a large part of global agriculture, often face challenges such as labour shortages and inefficient sowing techniques. As a result, productivity and sustainability are negatively impacted. Addressing these challenges is crucial to ensuring food security for the future.

Solution: The Seed Sowing Robot project focuses on automating the seed-sowing process in agriculture, aiming to reduce labour, minimize seed wastage, and increase crop yields through precise seed placement. The project involves assembling a robot using key electronic components, 3D printed parts, and laser-cut parts to build an affordable and effective solution.

Components Required:

- **<u>Arduino Uno:</u>**
- The Arduino Uno is a popular microcontroller board based on the ATmega328P chip.
- It has 14 digital input/output pins, 6 of which are analog input, which allow you to read sensor data.
- The board operates at 16 MHz and can be powered via a USB cable or an external battery.
- It's beginner-friendly, so you can experiment without much worry. If something goes wrong, the main chip can be easily replaced.

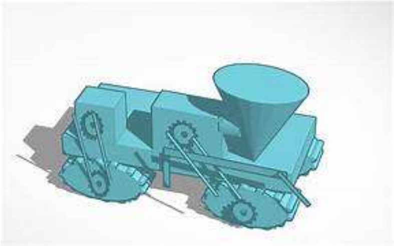
- **<u>Servo Motor (SKU#HK15138)</u>:** A servo motor (servomotor) is a highly specialized motor designed for precise control of rotary or linear motion. It's a rotational or translational motor that employs a feedback mechanism to ensure exact positioning, typically using a control signal that dictates the motor's movement to a desired position.

Ultrasonic Sensor (HC-SR04): An ultrasonic sensor is a device that uses sound waves to measure distance, speed, and position without making physical contact with an object. They work by sending out ultrasonic pulses and measuring the time it takes for the echo to return.

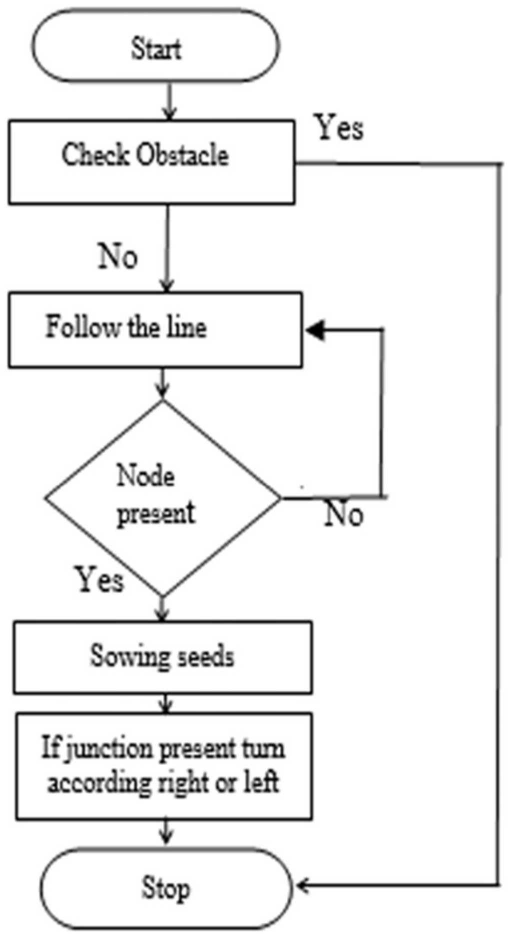
DC Motors with Encoders: DC motor encoders are used for speed control feedback in DC motors where an armature or rotor with wound wires rotates inside a magnetic field created by a stator. The DC motor encoder provides a mechanism to measure the speed of the rotor and provide closed-loop feedback to the drive for precise speed control.

Motor Driver (L298N): The L298N is a dual H-Bridge motor driver which allows speed and direction control of two DC motors at the same time. The module can drive DC motors that have voltages between 5 and 35V, with a peak current up to

Working:

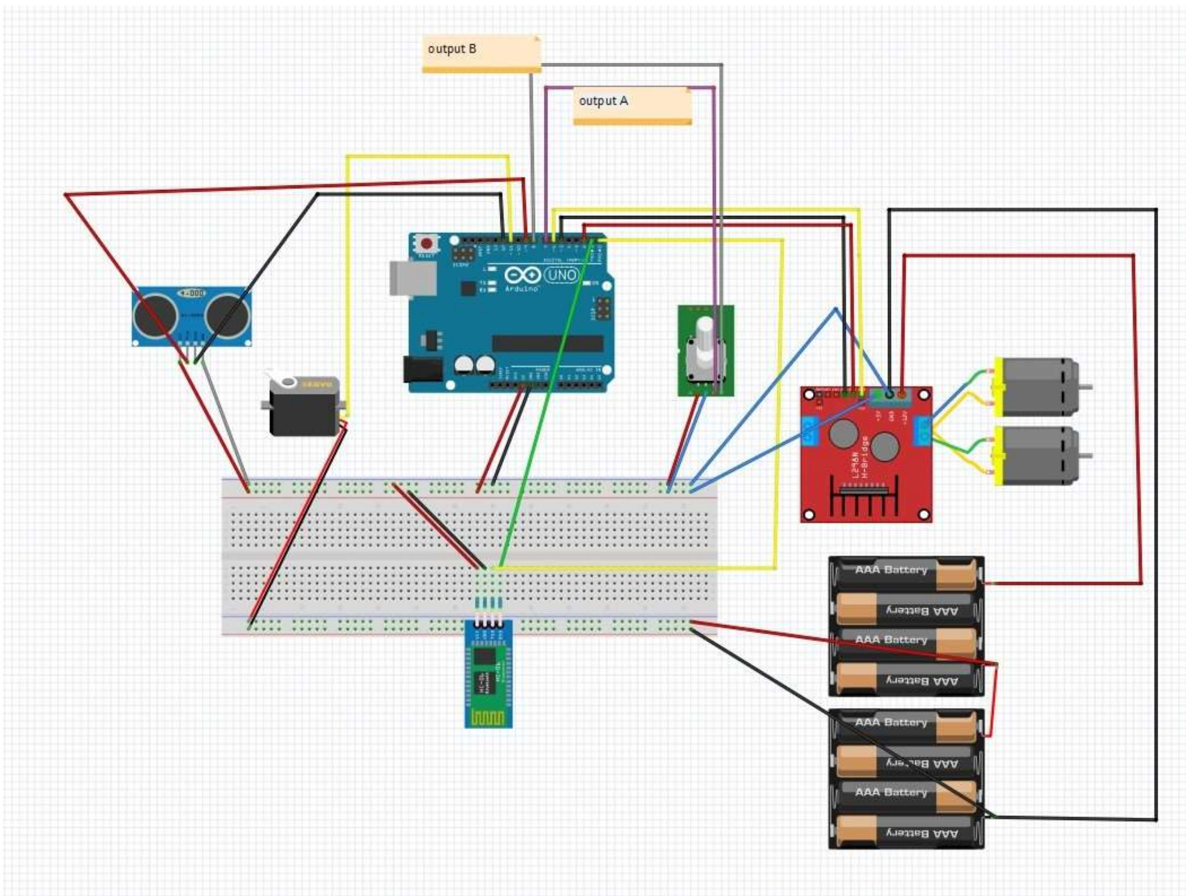


Automatic seed sowing robot works with the power supply from solar panel and battery. It requires the manual input which are of two types namely Vertical and Horizontal input. Vertical input refers to the distance between the two plants and horizontal input refers to the distance between the rows. The input is provided through once the input is given then the power switch is Switched ON. And power supply is provided to DC wheels. Based on the distance provided by farmer the robot moves to a fixed distance and stops. At the same time seed gear system rotates and seed falls through the pipe. Once when the robot stops the servo motor operates and makes the seed falling chamber open and after some delay it closes itself. Process continues and robot continues to move until the obstacle is found.



Once the obstacle is found it will be detected by ultrasonic sensor and it sends signal to the controller. As a result, the robot turns and moves certain distance which is given by the farmer (Horizontal input), it again turns back and continues to sow the seed. Here the robot turning is alternatively as it results in proper movement of robot through the field. Hence the system is designed fully automatic.

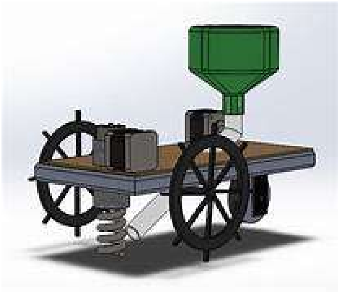
Circuit Design:



Future Enchancements and Expandability:

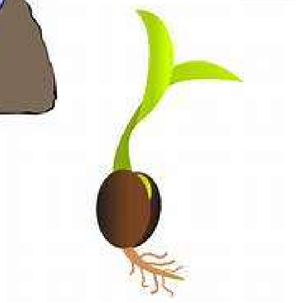
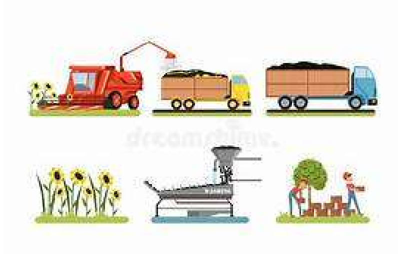
- Future enhancements to the Automatic Seed Sowing Machine using IoT could significantly elevate its capabilities and impact on modern agriculture. It uses real-time data analysis to optimize seed placement and timing based on environmental conditions and historical crop data.

- Variable rate seeding could also be introduced, allowing the machine to adjust seed rates automatically depending on soil fertility and moisture levels, thereby maximizing efficiency and crop yield. Additionally, enhancing the machine's navigation with autonomous capabilities, such as obstacle detection and avoidance, would enable it to operate independently in more complex field environments. These enhancements would collectively contribute to more efficient, precise, and sustainable farming practices.


NEVER UNDERESTIMATE
THE POWER OF A PLANTED
SEED.



Every seed
GROWS
into Something
AMAZING


THANK YOU !!



WHEN YOU
PLANT SEEDS
OF KINDNESS,
THEY BLOOM
ALL AROUND